



Dr. D. Y. Patil Educational Federation's  
**DR. D. Y. PATIL COLLEGE OF  
ENGINEERING & INNOVATION**  
Department of Artificial Intelligence and



**Data Science**

**Academic Year 2025-26**



**Academic Year: 2025-26**

## **LABORATORY MANUAL**

|                                                                          |                  |                       |
|--------------------------------------------------------------------------|------------------|-----------------------|
| <b>Name of the Student:</b>                                              |                  |                       |
| <b>Class: TE</b>                                                         | <b>Division:</b> | <b>Roll No.:</b>      |
| <b>Course Name: Software Laboratory III<br/>[ 317534 ] (2019 Course)</b> |                  | <b>Exam Seat No.:</b> |
| <b>Department of Artificial Intelligence and Data Science</b>            |                  |                       |

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### **Program Outcomes (PO's):**

POs are statements that describe what students are expected to know and be able to do upon graduating from the program. These relate to the skills, knowledge, analytical ability attitude and behavior that students acquire through the program.

• **PO1: Engineering Knowledge:**

Graduates will be able to apply the Knowledge of the mathematics, science and engineering fundamentals for the solution of engineering problems related to IT.

• **PO2: Problem Analysis:**

Graduates will be able to carry out identification and formulation of the problem statement by requirement engineering and literature survey.

• **PO3: Design/Development of Solutions:**

Graduates will be able to design a system, its components and/or processes to meet the required needs with consideration for public safety and social considerations.

• **PO4: Conduct Investigations of Complex Problems:**

Graduates will be able to investigate the problems, categorize the problem according to their complexity using modern computational concepts and tools.

• **PO5: Modern Tool Usage:**

Graduates will be able to use the techniques, skills, modern IT engineering tools necessary for engineering practice.

• **PO6: The Engineer and Society:**

Graduates will be able to apply reasoning and knowledge to assess global and societal issues

• **PO7: Environment and Sustainability:**

Graduates will be able to recognize the implications of engineering IT solution with respect to society and environment.

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• **PO8: Ethics:**

Graduates will be able to understand the professional and ethical responsibility.

• **PO9: Individual and Team Work:**

Graduates will be able to function effectively as an individual member, team member or leader in multi-disciplinary teams.

• **PO10: Communication:**

Graduates will be able to communicate effectively and make effective documentations and presentations.

• **PO11: Project Management and Finance:**

Graduates will be able to apply and demonstrate engineering and management principles in project management as a member or leader.

• **PO12: Life-long Learning:**

Graduates will be able to recognize the need for continuous learning and to engage in life-long learning.

- To understand principles of Data Science for the analysis of real time problems.
- To develop in depth understanding and implementation of the key technologies in Data Science and Data Analytics.
- To analyze and demonstrate knowledge of statistical data analysis techniques for decision-making.
- To gain practical, hands-on experience with statistics programming languages and Data tools.

### **Course Outcomes:**

*On completion of the course, students will be able to–*

**CO1:** Apply principles of Data Science for the analysis of real time problems

**CO2:** Implement data representation using statistical methods.

**CO3:** Implement and evaluate data analytics algorithms.

**CO4:** Perform text preprocessing.

**CO5:** Implement data visualization techniques.

**CO6:** Use cutting edge tools and technologies to analyze Data.

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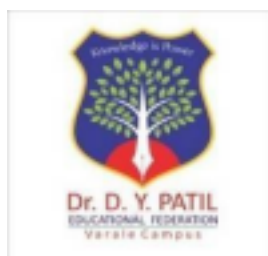
**Artificial Intelligence and Data Science  
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# Department of Artificial Intelligence and Data Science

Academic Year 2025-26



## CERTIFICATE

This is to certify that Mr. /Ms. \_\_\_\_\_ of Class TE - AI-DS, Roll No. \_\_\_\_\_ Examination Seat No. \_\_\_\_\_ has completed all the practical work in the Software Laboratory - III [317534] satisfactorily, as prescribed by Savitribai Phule Pune University, Pune in the academic year 2025-26 (Term-II).

Place:

Date:

Course In-charge HOD Principal Department of AI-DS Department of AI-DS DYPCOEL, Varale

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Academic Year: 2025-26 Department: AI-DS

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|---------|------------------------|--------------------|------------------|----------|------|--------|
|---------|------------------------|--------------------|------------------|----------|------|--------|

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |  |  |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 1 | <p><b>Assignment: Data Wrangling I</b></p> <p>Perform the following operations using Python on any open source dataset (e.g., data.csv)</p> <p>Import all the required Python Libraries.</p> <ol style="list-style-type: none"> <li>1. Locate open source data from the web.</li> <li>2. Provide a clear description of the data and its source.</li> <li>3. Load the Dataset into the pandas data frame.</li> <li>4. Data Pre-processing:</li> <li>5. Data Formatting and Data Normalization:</li> <li>6. Turn categorical variables into quantitative variables in Python.</li> </ol> |  |  |  |  |  |
| 2 | <p><b>Assignment: Data Wrangling, II</b></p> <p>Create an “Academic performance” dataset of students and perform the following operations using Python.</p> <ol style="list-style-type: none"> <li>1. Scan all variables for missing values and inconsistencies. If there are missing values use any of the suitable techniques to deal with them.</li> <li>2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.</li> <li>3. Apply data transformations on at least one of the variables.</li> </ol>                |  |  |  |  |  |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |  |  |  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 3 | <p><b>Assignment: Descriptive Statistics - Measures of Central Tendency and variability</b></p> <p>Perform the following operations on any open source dataset (e.g., data.csv)</p> <ol style="list-style-type: none"> <li>1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative variables.</li> <li>2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris versicolor' and 'Iris- versicolor' of iris.csv dataset.</li> </ol> |  |  |  |  |  |
| 4 | <p><b>Assignment: Data Analytics I</b></p> <p>Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset. The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.</p>                                                                                                                                                                                                                                                                                                                      |  |  |  |  |  |
| 5 | <p><b>Assignment: Data Analytics II</b></p> <ol style="list-style-type: none"> <li>1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset.</li> <li>2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.</li> </ol>                                                                                                                                                                                                                                                                                                                    |  |  |  |  |  |
| 6 | <p><b>Assignment: Data Analytics III</b></p> <ol style="list-style-type: none"> <li>1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  |  |  |



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|   |                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  |  |  |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|   | 2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.                                                                                                                                                                                                                                                                                                        |  |  |  |  |  |
| 7 | <b>Assignment: Text Analytics</b><br>1. Extract Sample document and apply following document preprocessing methods: Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization.<br>2. Create representation of documents by calculating Term Frequency and Inverse Document Frequency.                                                                                                                    |  |  |  |  |  |
| 8 | <b>Assignment: Data Visualization I 1.</b><br>Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship. Use the Seaborne library to see if we can find any patterns in the data.<br>2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram. |  |  |  |  |  |
| 9 | <b>Assignment: Data Visualization II 1.</b><br>Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. 2. Write observations on the inference from the above statistics.                                                                                                      |  |  |  |  |  |



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|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>10</b> | <b>Assignment: Data Visualization III</b><br>Download the Iris flower dataset or any other dataset into a Data Frame. (e.g., <a href="https://archive.ics.uci.edu/ml/datasets/Iris">https://archive.ics.uci.edu/ml/datasets/Iris</a> ). Scan the dataset and give the inference as:<br>1. List down the features and their types (e.g., numeric, nominal) available in the dataset.<br>2. Create a histogram for each feature in the dataset to illustrate the feature distributions.<br>3. Create a boxplot for each feature in the dataset.<br>4. Compare distributions and identify outliers. |  |  |  |  |  |
| <b>11</b> | <b>Assignment:</b><br>Create databases and tables, insert small amounts of data, and run simple queries using Impala                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |  |  |
| <b>12</b> | <b>Assignment:</b><br>Design a distributed application using MapReduce which processes a log file of a system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |  |  |
| <b>13</b> | <b>Assignment:</b><br>Write a simple program in SCALA using Apache Spark framework                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |  |  |
| <b>14</b> | Mini Projects/ Case Study PYTHON/R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |  |  |

**Name & Signature of Course In-charge**



## **Experiment No: 1**

### Data Wrangling- I

**Name of the Student:**

\_\_\_\_\_ **Class: Roll No.:**

**Batch:**

Date: Mark: /10

**Signature of the Course In-charge:** \_\_\_\_\_

**Signature of the HOD:**

\_\_\_\_\_

**Group A**  
**Assignment No: 1**

**Title of the Assignment: Data Wrangling- I**

Perform the following operations using Python on any open source dataset (e.g., data.csv). Import all the required Python Libraries.

1. Locate open source data from the web (e.g. <https://www.kaggle.com>).
2. Provide a clear description of the data and its source (i.e., URL of the web site).
3. Load the Dataset into the panda's data frame.
4. Data Preprocessing: check for missing values in the data using pandas isnull(), describe(), function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame.
5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.
6. Turn categorical variables into quantitative variables in Python.

**Objective:** To perform the data wrangling operation using Python on any open source dataset

**Prerequisite:**

1. Basic of Python Programming
2. Concept of Data Preprocessing, Data Formatting, Data Normalization and Data Cleaning.

**Software Requirements:**

- Anaconda with Python 3.7

**Hardware Requirement:**

- PIV, 2GB RAM, 500 GB HDD.



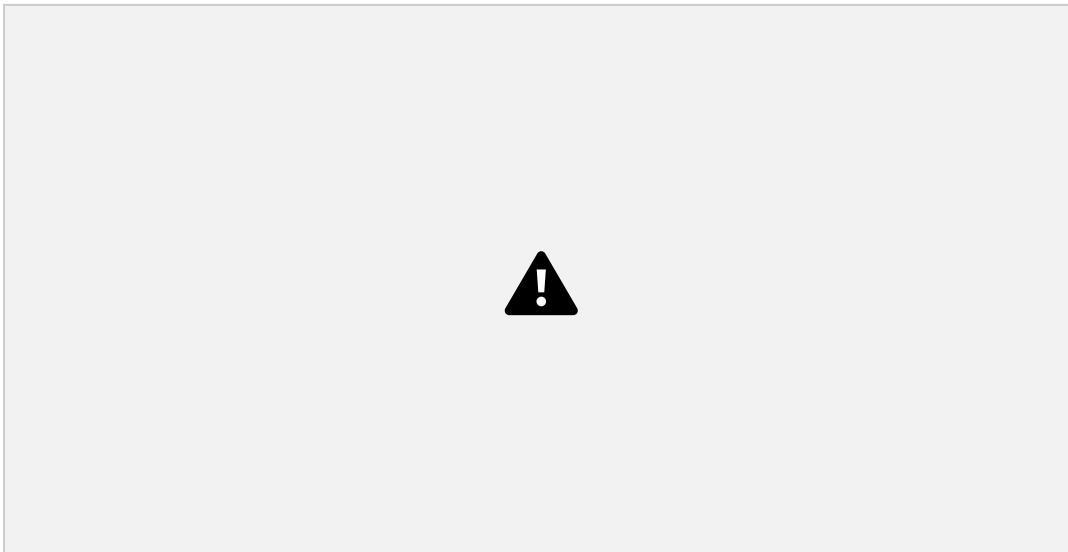
**Outcome:**

Students will be able to locate and load open-source datasets into Python using Pandas. They will understand data pre-processing steps such as handling missing values, checking statistics, and identifying data types. Students will learn to format and normalize data by correcting variable types and finally convert categorical variables into numerical representations for analysis.

**Theory:**

**1. Introduction to Dataset**

A dataset is a collection of records, similar to a relational database table. Records are similar to table rows, but the columns can contain not only strings or numbers, but also nested data structures such as lists, maps, and other records.



**Instance:** A single row of data is called an instance. It is an observation from the domain.

**Feature:** A single column of data is called a feature. It is a component of an observation and is also called an attribute of a data instance. Some features may be inputs to a model (the predictors) and others may be outputs or the features to be predicted.

**Data Type:** Features have a data type. They may be real or integer-valued or may have a categorical or ordinal value. You can have strings, dates, times, and more complex types, but typically they are reduced to real or categorical values when working with traditional machine learning methods.

**Datasets:** A collection of instances is a dataset and when working with machine learning methods we typically need a few datasets for different purposes.

**Training Dataset:** A dataset that we feed into our machine learning algorithm to train our model.

**Testing Dataset:** A dataset that we use to validate the accuracy of our model but is not used to train the model. It may be called the validation dataset.

### **Data Represented in a Table:**

Data should be arranged in a two-dimensional space made up of rows and columns. This type of data structure makes it easy to understand the data and pinpoint any problems. An example of some raw data stored as a CSV (comma separated values).



The representation of the same data in a table is as follows:



### **Pandas Data Types:**

A data type is essentially an internal construct that a programming language uses to understand how to store and manipulate data. A possible confusing point about pandas data types is that there is some overlap between pandas, python and numpy. This table summarizes the key



| Pandas type   | Python type  | NumPy type                                                     | Usage                                        |
|---------------|--------------|----------------------------------------------------------------|----------------------------------------------|
| object        | str or mixed | string_, unicode_, mixed types                                 | Text or mixed numeric and non-numeric values |
| int64         | int          | int_, int8, int16, int32, int64, uint8, uint16, uint32, uint64 | Integer numbers                              |
| float64       | float        | float_, float16, float32, float64                              | Floating point numbers                       |
| bool          | bool         | bool_                                                          | True/False values                            |
| datetime64    | NA           | datetime64[ns]                                                 | Date and time values                         |
| timedelta[ns] | NA           | NA                                                             | Differences between two datetimes            |
| category      | NA           | NA                                                             | Finite list of text values                   |

## 2. Python Libraries for Data Science

### a. Pandas

Pandas is an open-source Python package that provides high-performance, easy-to-use data structures and data analysis tools for the labeled data in Python programming language. **What can you do with Pandas?**

1. Indexing, manipulating, renaming, sorting, merging data frame
2. Update, Add, Delete columns from a data frame
3. Impute missing files, handle missing data or NaNs
4. Plot data with histogram or box plot

### b. NumPy

One of the most fundamental packages in Python, NumPy is a general-purpose array processing package. It provides high-performance multidimensional array objects and tools to work with the arrays. NumPy is an efficient container of generic multi-dimensional data. NumPy's main object is the homogeneous multidimensional array. It is a table of elements or numbers of the same datatype, indexed by a tuple of positive integers. In NumPy, dimensions are called axes and the number of axes is called rank. NumPy's array class is called ndarray aka array.



### **What can you do with NumPy?**

1. Basic array operations: add, multiply, slice, flatten, reshape, index arrays
2. Advanced array operations: stack arrays, split into sections, broadcast arrays
3. Work with DateTime or Linear Algebra
4. Basic Slicing and Advanced Indexing in NumPy Python

### **c. Matplotlib**

This is undoubtedly my favorite and a quintessential Python library. You can create stories with the data visualized with Matplotlib. Another library from the SciPy Stack, Matplotlib plots 2D figures.

#### **What can you do with Matplotlib?**

Histogram, bar plots, scatter plots, area plot to pie plot, Matplotlib can depict a wide range of visualizations. With a bit of effort and tint of visualization capabilities, with Matplotlib, you can create just any visualizations: Line plots

- Scatter plots
- Area plots
- Bar charts and Histograms
- Pie charts
- Stem plots
- Contour plots
- Quiver plots
- Spectrograms

Matplotlib also facilitates labels, grids, legends, and some more formatting entities with Matplotlib.

### **d. Seaborn**

So when you read the official documentation on Seaborn, it is defined as the data visualization library based on Matplotlib that provides a high-level interface for drawing attractive and informative statistical graphics. Putting it simply, seaborn is an extension of Matplotlib with advanced features.



**What can you do with Seaborn?**

1. Determine relationships between multiple variables (correlation)
2. Observe categorical variables for aggregate statistics
3. Analyze univariate or bi-variate distributions and compare them between different data subsets
4. Plot linear regression models for dependent variables
5. Provide high-level abstractions, multi-plot grids
6. Seaborn is a great second-hand for R visualization libraries like corrplot and ggplot.

**e. Scikit Learn**

Introduced to the world as a Google Summer of Code project, Scikit Learn is a robust machine learning library for Python. It features ML algorithms like SVMs, random forests, k-means clustering, spectral clustering, mean shift, cross-validation and more... Even NumPy, SciPy and related scientific operations are supported by Scikit Learn with Scikit Learn being a part of the SciPy Stack.

**What can you do with Scikit Learn?**

1. Classification: Spam detection, image recognition
2. Clustering: Drug response, Stock price
3. Regression: Customer segmentation, Grouping experiment outcomes
4. Dimensionality reduction: Visualization, Increased efficiency
5. Model selection: Improved accuracy via parameter tuning
6. Pre-processing: Preparing input data as a text for processing with machine learning algorithms.

**3. Description of Dataset:**

The Iris dataset was used in R.A. Fisher's classic 1936 paper, The Use of Multiple Measurements in Taxonomic Problems, and can also be found on the UCI Machine Learning Repository. It includes three iris species with 50 samples each as well as some properties about each flower. One flower species is linearly separable from the other two, but the other two are not linearly separable from each other.



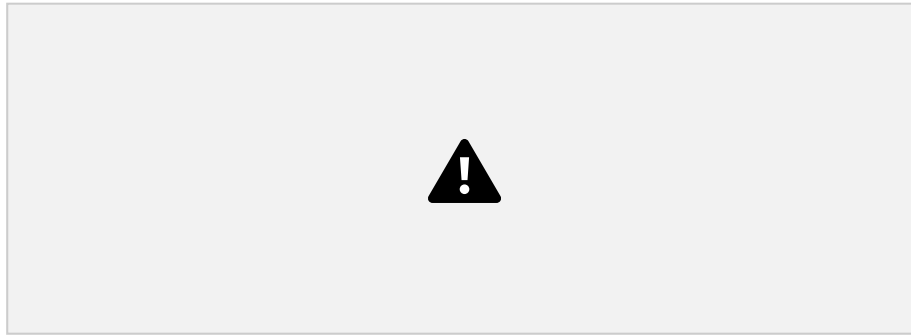


**Total Sample- 150**

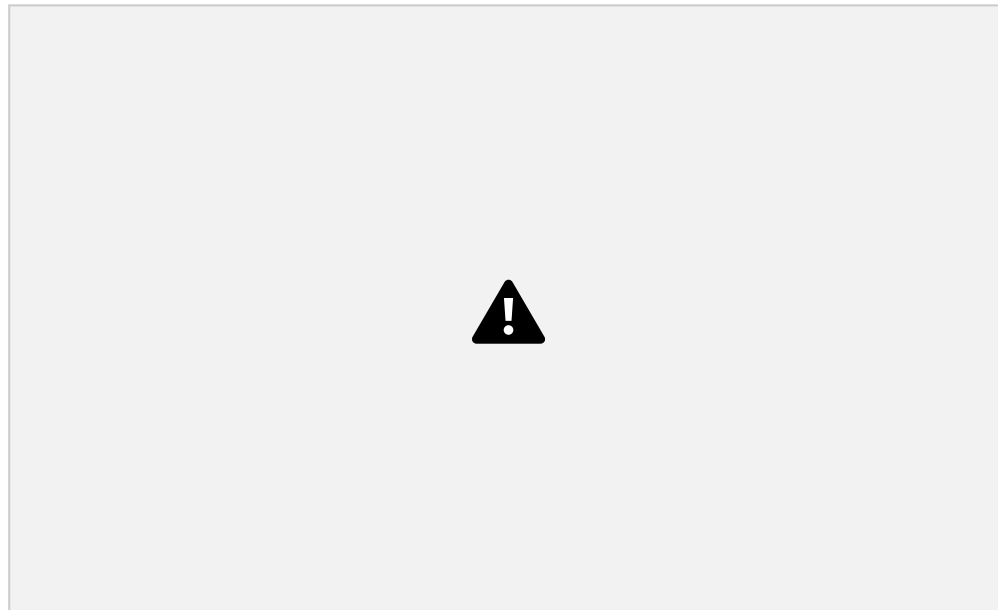
**The columns in this dataset are:**

1. Id
2. SepalLengthCm
3. SepalWidthCm
4. PetalLengthCm
5. PetalWidthCm
6. Species

**3 Different Types of Species each contain 50 Sample**



**Description of Dataset-**



#### 4. Panda Dataframe functions for Load Dataset

# The columns of the resulting DataFrame have different dtypes. iris.dtypes

1. The dataset is downloads from UCI repository.

**csv\_url='https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'**

2. Now Read CSV File as a Dataframe in Python from from path where you saved the same The Iris data set is stored in .csv format. '.csv' stands for comma separated values. It is easier to load .csv files in Pandas data frame and perform various analytical operations on it.

Load Iris.csv into a Pandas data frame —

##### **Syntax**

**iris = pd.read\_csv(csv\_url, header = None)**

3. The csv file at the UCI repository does not contain the variable/column names. They are located in a separate file.

**col\_names = ['Sepal\_Length','Sepal\_Width','Petal\_Length','Petal\_Width','Species']**

4. read in the dataset from the UCI Machine Learning Repository link and specify column names to use

**iris = pd.read\_csv(csv\_url, names = col\_names)**



### 5. Panda Data Frame functions for Data Preprocessing: Data frame Operations:

| Sr. No. | Data Frame Function                                       | Description                                                                                                                                                                                                                                            |
|---------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <b>dataset.head(n=5)</b>                                  | <b>Return the first n rows.</b>                                                                                                                                                                                                                        |
| 2       | <b>dataset.tail(n=5)</b>                                  | <b>Return the last n rows.</b>                                                                                                                                                                                                                         |
| 3       | <b>dataset.index</b>                                      | The index (row labels) of the Dataset.                                                                                                                                                                                                                 |
| 4       | <b>dataset.columns</b>                                    | The column labels of the Dataset.                                                                                                                                                                                                                      |
| 5       | <b>dataset.shape</b>                                      | Return a tuple representing the dimensionality of the Dataset.                                                                                                                                                                                         |
| 6       | <b>dataset.dtypes</b>                                     | Return the dtypes in the Dataset.<br>This returns a Series with the data type of each column. The result's index is the original Dataset's columns. Columns with mixed types are stored with the object dtype.                                         |
| 7       | <b>dataset.columns.values</b>                             | Return the columns values in the Dataset in array format                                                                                                                                                                                               |
| 8       | <b>dataset.describe(include='all')</b>                    | Generate descriptive statistics. to view some basic statistical details like percentile, mean, std etc. of a data frame or a series of numeric values.<br>Analyzes both numeric and object series, as well as Dataset column sets of mixed data types. |
| 9       | <b>dataset['Column name']</b>                             | Read the Data Column wise.                                                                                                                                                                                                                             |
| 10      | <b>dataset.sort_index(axis=1, ascending=False)</b>        | Sort object by labels (along an axis).                                                                                                                                                                                                                 |
| 11      | <b>dataset.sort_values(by="Column name")</b>              | Sort values by column name.                                                                                                                                                                                                                            |
| 12      | <b>dataset.iloc[5]</b>                                    | Purely integer-location based indexing for selection by position.                                                                                                                                                                                      |
| 13      | <b>dataset[0:3]</b>                                       | Selecting via [], which slices the rows.                                                                                                                                                                                                               |
| 14      | <b>dataset.loc[:,<br/>["Col_name1",<br/>"col_name2"]]</b> | Selection by label                                                                                                                                                                                                                                     |
| 15      | <b>dataset.iloc[:n, :]</b>                                | a subset of the first n rows of the original data                                                                                                                                                                                                      |
| 16      | <b>dataset.iloc[:, :n]</b>                                | a subset of the first n columns of the original data                                                                                                                                                                                                   |
| 17      | <b>dataset.iloc[:m, :n]</b>                               | a subset of the first m rows and the first n columns                                                                                                                                                                                                   |



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#### Few Examples of iLoc to slice data for iris Dataset

| Sr. No | Data Frame Function                           | Description                                                                | Output                                                                                |
|--------|-----------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1      | <code>dataset.iloc[3:5, 0:2]</code>           | Slice the data                                                             |    |
| 2      | <code>dataset.iloc[[1, 2, 4], [0, 2]]</code>  | By lists of integer position locations, similar to the NumPy/Python style: |    |
| 3      | <code>dataset.iloc[1:3, :]</code>             | For slicing rows explicitly:                                               |  |
| 4      | <code>dataset.iloc[:, 1:3]</code>             | For slicing Column explicitly:                                             |  |
| 4      | <code>dataset.iloc[1, 1]</code>               | For getting a value explicitly:                                            |  |
| 5      | <code>dataset['SepalLengthCm'].iloc[5]</code> | Accessing Column and Rows by position                                      |  |

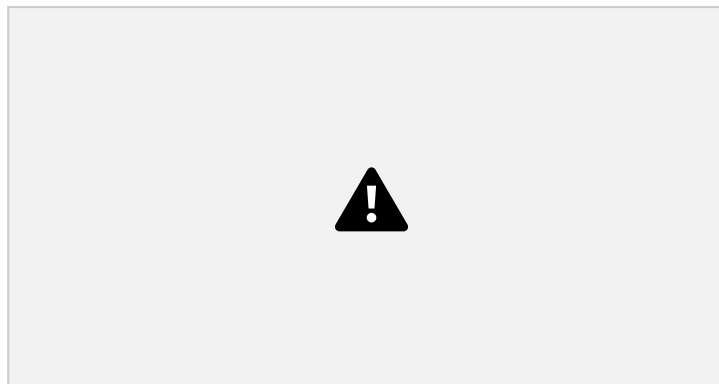


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|   |                                                                   |                                                              |                                                                                     |
|---|-------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 6 | <b>cols_2_4=dataset<br/>t.c olumns[2:4]<br/>dataset[cols_2_4]</b> | Get Column<br>Name then get<br>data from<br>column           |  |
| 7 | <b>dataset[dataset.col<br/>umns[2:4]].iloc[5:<br/>1 0]</b>        | in one Expression<br>answer for the<br>above two<br>commands |  |

### Checking of Missing Values in Dataset:

- **isnull()** is the function that is used to check missing values or null values in pandas python. • **isna()** function is also used to get the count of missing values of column and row wise count of missing values
- The dataset considered for explanation is:



a. is there any missing values in dataframe as a whole

**Function:** DataFrame.isnull()

**Output:**



a. is there any missing values across each

**column Function:** `DataFrame.isnull().any()`

**Output:**



b. **count of missing values across each column using isna() and isnull()** In order to get the count of missing values of the entire dataframe `isnull()` function is used. `sum()` which does the column wise sum first and doing another `sum()` will get the count of missing values of the entire dataframe.

**Function:** `dataframe.isnull().sum().sum()`

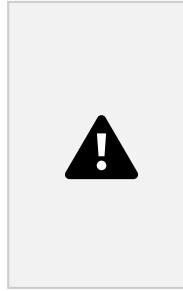
**Output :** 8

c. **count row wise missing value using**

**isnull() Function:**

`dataframe.isnull().sum(axis = 1)` **Output:**





**d. count Column wise missing value using isnull()**

**Method 1:**

**Function:** dataframe.isnull().sum()

**Output:**



**Method 2:**

**unction:** dataframe.isna().sum()



**e. count of missing values of a specific column.**

**Function:** dataframe.col\_name.isnull().sum()

df1.Gender.isnull().sum()

**Output: 2**

**f. groupby count of missing values of a column.**

In order to get the count of missing values of the particular column by group in pandas we will be using isnull() and sum() function with apply() and groupby() which performs the group wise count of missing values as shown below.

**Function:** df1.groupby(['Gender'])['Score'].apply(lambda x: x.isnull().sum())



## Output:



### 6. Panda functions for Data Formatting and Normalization

The Transforming data stage is about converting the data set into a format that can be analyzed or modelled effectively, and there are several techniques for this process.

- a. **Data Formatting:** Ensuring all data formats are correct (e.g. object, text, floating number, integer, etc.) is another part of this initial ‘cleaning’ process. If you are working with dates in Pandas, they also need to be stored in the exact format to use special date-time functions.

#### Functions used for Data Formatting

| Sr. No | Data Frame Function                                                   | Description                                                              | Output |
|--------|-----------------------------------------------------------------------|--------------------------------------------------------------------------|--------|
| 1.     | <b>df.dtypes</b>                                                      | To check the data type                                                   |        |
| 2.     | <b>df['petal length (cm)']= df['petal length (cm)'].astype("int")</b> | To change the data type (data type of ‘petal length (cm)’changed to int) |        |

- a. **Data normalization:** Mapping all the nominal data values onto a uniform scale (e.g. from 0 to 1) is involved in data normalization. Making the ranges consistent across variables





helps with statistical analysis and ensures better comparisons later on. It is also known as Min-Max scaling.

**Algorithm:**

**Step 1 :** Import pandas and sklearn library for preprocessing

```
from sklearn import preprocessing
```

**Step 2:** Load the iris dataset in dataframe object df

**Step 3:** Print iris dataset.

```
df.head()
```

**Step 5:** Create a minimum and maximum processor object

```
min_max_scaler = preprocessing.MinMaxScaler()
```

**Step 6:** Separate the feature from the class label

```
x=df.iloc[:,4]
```

**Step 6:** Create an object to transform the data to fit minmax processor

```
x_scaled = min_max_scaler.fit_transform(x)
```

**Step 7:** Run the normalizer on the dataframe

```
df_normalized = pd.DataFrame(x_scaled)
```

**Step 8:** View the dataframe

```
df_normalized
```

**Output: After Step 3:**



**Output after step 8:**



## **7. Panda Functions for handling categorical variables**

- **Categorical variables** have values that **describe a ‘quality’ or ‘characteristic’** of a data unit, like **‘what type’ or ‘which category’**.
- Categorical variables fall into **mutually exclusive (in one category or in another)** and **exhaustive (include all possible options)** categories. Therefore, categorical variables are qualitative variables and **tend to be represented by a non-numeric value**.
- Categorical features refer **to string type data** and can be easily understood by human beings. But in case of a **machine, it cannot interpret the categorical data directly**. Therefore, the categorical data must be **translated into numerical data that can be understood by machine**.

There are many ways to convert categorical data into numerical data. Here the three most used methods are discussed.

- a. **Label Encoding:** Label Encoding refers to **converting the labels into a numeric form** so as to convert them into the machine-readable form. **It is an important preprocessing step for the structured dataset** in supervised learning.



**b. Example:** Suppose we have a column Height in some dataset. After applying label encoding, the Height column is converted into:



where 0 is the label for tall, 1 is the label for medium, and 2 is a label for short height. **Label Encoding on iris dataset:** For iris dataset the target column which is Species. It contains three species Iris-setosa, Iris-versicolor, Iris-virginica.

#### **Sklearn Functions for Label Encoding:**

- **preprocessing.LabelEncoder:** It Encode labels with value between 0 and n\_classes - 1.
- **fit\_transform(y):**

**Parameters:** yarray-like of shape (n\_samples,) **Target values.**

**Returns:** yarray-like of shape (n\_samples,)

**Encoded labels.**

This transformer should be used to encode target values, and not the input.

#### **Algorithm:**

**Step 1 :** Import pandas and sklearn library for preprocessing

```
from sklearn import preprocessing
```

**Step 2:** Load the iris dataset in dataframe object df

**Step 3:** Observe the unique values for the Species column.

```
df['Species'].unique()
```

**output:** array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)

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```
label_encoder = preprocessing.LabelEncoder()
```

**Step 5:** Encode labels in column 'species'.

```
df['Species']= label_encoder.fit_transform(df['Species'])
```

**Step 6:** Observe the unique values for the Species column.

```
df['Species'].unique()
```

**Output:** array([0, 1, 2], dtype=int64)

- Use LabelEncoder when there are only two possible values of a categorical feature. For example, features having value such as yes or no. Or, maybe, gender features when there are only two possible values including male or female.

**Limitation:** Label encoding converts the data in machine-readable form, but it assigns a **unique number(starting from 0) to each class of data**. This may lead to the generation of **priority issues in the data sets**. A label with a high value may be considered to have high priority than a label having a lower value.

### c. One-Hot Encoding:

In one-hot encoding, we create a new set of dummy (binary) variables that is equal to the number of categories (k) in the variable. For example, let's say we have a categorical variable Color with three categories called "Red", "Green" and "Blue", we need to use three dummy variables to encode this variable using one-hot encoding. A dummy (binary) variable just takes the value 0 or 1 to indicate the exclusion or inclusion of a category.



“Red” color is encoded as  $[1 \ 0 \ 0]$  vector of size 3.

“Green” color is encoded as  $[0 \ 1 \ 0]$  vector of size 3.

“Blue” color is encoded as  $[0 \ 0 \ 1]$  vector of size 3.

**One-hot encoding on iris dataset:** For iris dataset the target column which is Species. It contains three species Iris-setosa, Iris-versicolor, Iris-virginica.

### Sklearn Functions for One-hot Encoding:

- **sklearn.preprocessing.OneHotEncoder():** Encode categorical integer features using a one-hot aka one-of-K scheme

### Algorithm:

**Step 1 :** Import pandas and sklearn library for preprocessing  
from sklearn import preprocessing

**Step 2:** Load the iris dataset in dataframe object df

**Step 3:** Observe the unique values for the Species column.

```
df['Species'].unique()
```

**output:** array(['Iris-setosa', 'Iris-versicolor', 'Iris virginica'],  
dtype=object)

**Step 4:** Apply label\_encoder object for label encoding the Observe the unique values for the Species column.

```
df['Species'].unique()
```

**Output:** array([0, 1, 2], dtype=int64)

**Step 5:** Remove the target variable from dataset

```
features_df=df.drop(columns=['Species'])
```

**Step 6:** Apply one\_hot encoder for Species column.

```
enc = preprocessing.OneHotEncoder()
```

```
enc_df=pd.DataFrame(enc.fit_transform(df[['Species']])).toarray()
```

**Step 7:** Join the encoded values with Features variable

```
df_encode = features_df.join(enc_df)
```

**Step 8:** Observe the merge dataframe

df\_encode

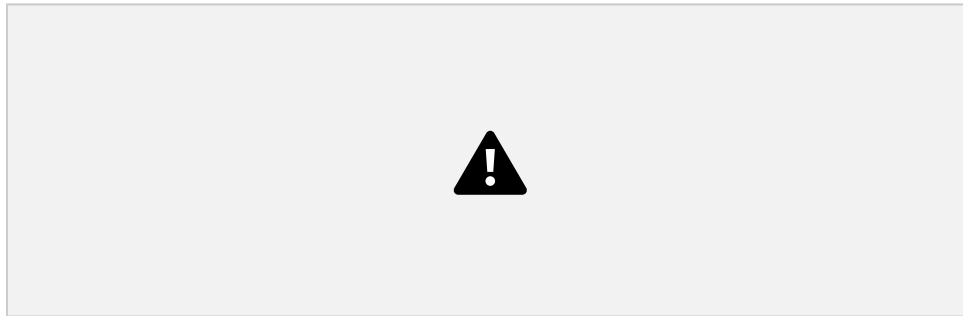
**Step 9:** Rename the newly encoded columns.

```
df_encode.rename(columns = {0:'Iris-Setosa',  
1:'Iris-Versicolor',2:'Iris-virginica'}, inplace = True)
```

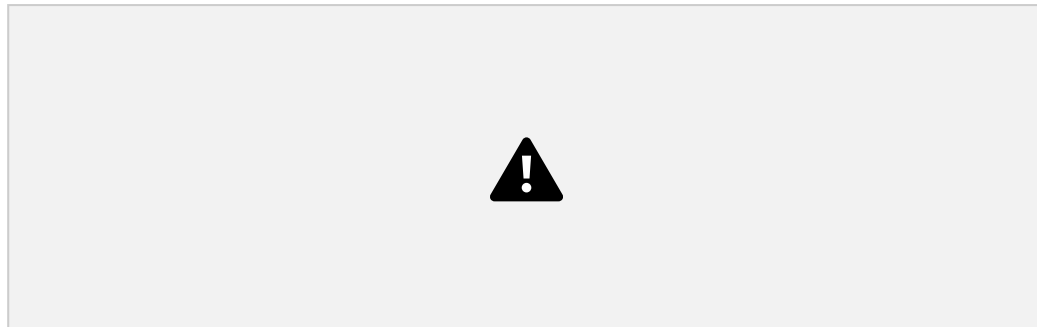
**Step 10:** Observe the merge dataframe

df\_encode

**Output after Step 8:**



**Output after Step 10:**



#### **a. Dummy Variable Encoding**

Dummy encoding also uses dummy (binary) variables. Instead of creating a number of dummy variables that is equal to the number of categories (k) in the variable, dummy encoding uses k-1 dummy variables. To encode the same Color variable with three categories using the dummy encoding, we need to use only two dummy variables.





In dummy encoding,

“Red” color is encoded as  $[1\ 0]$  vector of size 2.

“Green” color is encoded as  $[0\ 1]$  vector of size 2.

“Blue” color is encoded as  $[0\ 0]$  vector of size 2.

Dummy encoding removes a duplicate category present in the one-hot encoding.

**Pandas Functions for One-hot Encoding with dummy variables: •**

**pandas.get\_dummies(data, prefix=None, prefix\_sep='\_', dummy\_na=False, columns=None, sparse=False, drop\_first=False, dtype=None):** Convert categorical variable into dummy/indicator variables.

• **Parameters:**

**data:** array-like, Series, or DataFrame

Data of which to get dummy indicators.

**prefixstr:** list of str, or dict of str, default None

String to append DataFrame column names.

**prefix\_sep:** str, default ‘\_’

If appending prefix, separator/delimiter to use. Or pass a list or dictionary as with prefix.

**dummy\_na:** bool, default False

Add a column to indicate NaNs, if False NaNs are ignored.

**columns:** list-like, default None

Column names in the DataFrame to be encoded. If columns is None then all the columns with object or category dtype will be converted.



**sparse: bool: default False**

Whether the dummy-encoded columns should be backed by a SparseArray (True) or a regular NumPy array (False).

**drop\_first: bool, default False**

Whether to get k-1 dummies out of k categorical levels by removing the first level.

**dtype: dtype, default np.uint8**

Data type for new columns. Only a single dtype is allowed.

- **Return :** DataFrame with Dummy-coded data.

### Algorithm:

**Step 1 :** Import pandas and sklearn library for preprocessing

from sklearn import preprocessing

**Step 2:** Load the iris dataset in dataframe object df

**Step 3:** Observe the unique values for the Species column.

df['Species'].unique()

**output:** array(['Iris-setosa', 'Iris-versicolor', 'Iris virginica'], dtype=object)

**Step 4:** Apply label\_encoder object for label encoding the Observe the unique values for the Species column.

df['Species'].unique()

Output: array([0, 1, 2], dtype=int64)

**Step 6:** Apply one\_hot encoder with dummy variables for Species column. one\_hot\_df = pd.get\_dummies(df, prefix="Species", columns=['Species'], drop\_first=True) **Step**

**7:** Observe the merge dataframe

one\_hot\_df







**Conclusion-** In this way we have explored the functions of the python library for Data Preprocessing, Data Wrangling Techniques and How to Handle missing values on Iris Dataset.

**Viva Questions:**

1. Explain Data Frame with Suitable example.
2. What is the limitation of the label encoding method?
3. What is the need of data normalization?
4. What are the different Techniques for Handling the Missing Data?

**Coding**

**Completion**

**Total Dated Sign of Course In-charge**

**Efficiency Viva Timely**

**5 3 2 10**



## **Experiment No: 2**

### **Data Wrangling- II**

**Name of the Student:**

\_\_\_\_\_ **Class: Roll No.:**

**Batch:**

Date: Mark: /10

**Signature of the Course In-charge:** \_\_\_\_\_

**Signature of the HOD:**

\_\_\_\_\_



**Group A**

**Assignment No: 2**

**Title of the Assignment: Data Wrangling, II**

Create an “Academic performance” dataset of students and perform the following operations using Python.

1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.
3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution.

**Objective:** To perform the data wrangling operation using Python on any open source dataset

**Prerequisite:**

1. Basic of Python Programming
2. Concept of Data Preprocessing, Data Formatting, Data Normalization and Data Cleaning.

**Software Requirements:**

- Anaconda with Python 3.7

**Hardware Requirement:**

- PIV, 2GB RAM, 500 GB HDD.

**Outcome:**

Students understand how to identify and handle missing values, inconsistencies, and outliers in an academic performance dataset. It also enables them to apply suitable data transformation techniques to improve data quality, interpretability, and model performance.



## Theory:

### 1. Creation of Dataset using Microsoft Excel.

The dataset is created in “CSV” format.

- The name of dataset is **StudentsPerformance**
- **The features of the dataset are:** Math\_Score, Reading\_Score, Writing\_Score, Placement\_Score, Club\_Join\_Date .
- **Number of Instances:** 30
- **The response variable is:** Placement\_Offer\_Count .
- **Range of Values:**  
Math\_Score [60-80], Reading\_Score[75-95], ,Writing\_Score [60,80],  
Placement\_Score[75-100], Club\_Join\_Date [2018-2021].
- **The response variable is** the number of placement offers facilitated to particular students, which is largely depend on Placement\_Score

To fill the values in the dataset the **RANDBETWEEN** is used. Returns a random integer number between the numbers you specify

**Syntax: RANDBETWEEN(bottom, top)** **Bottom** The smallest integer and

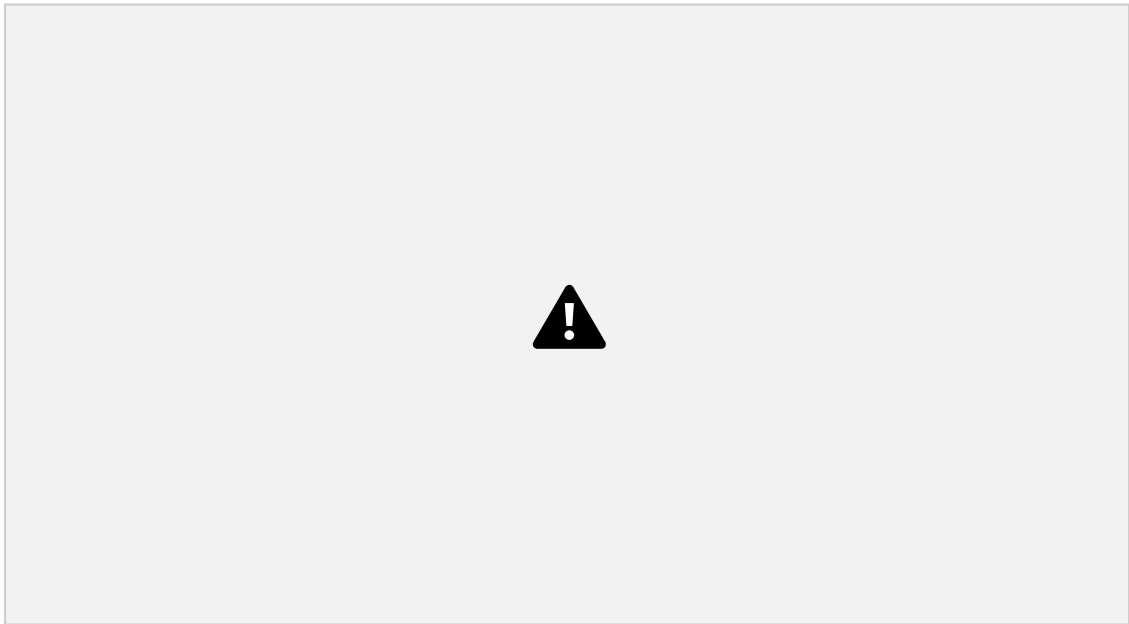
**Top** The largest integer RANDBETWEEN will return.

For better understanding and visualization, 20% impurities are added into each variable to the dataset.



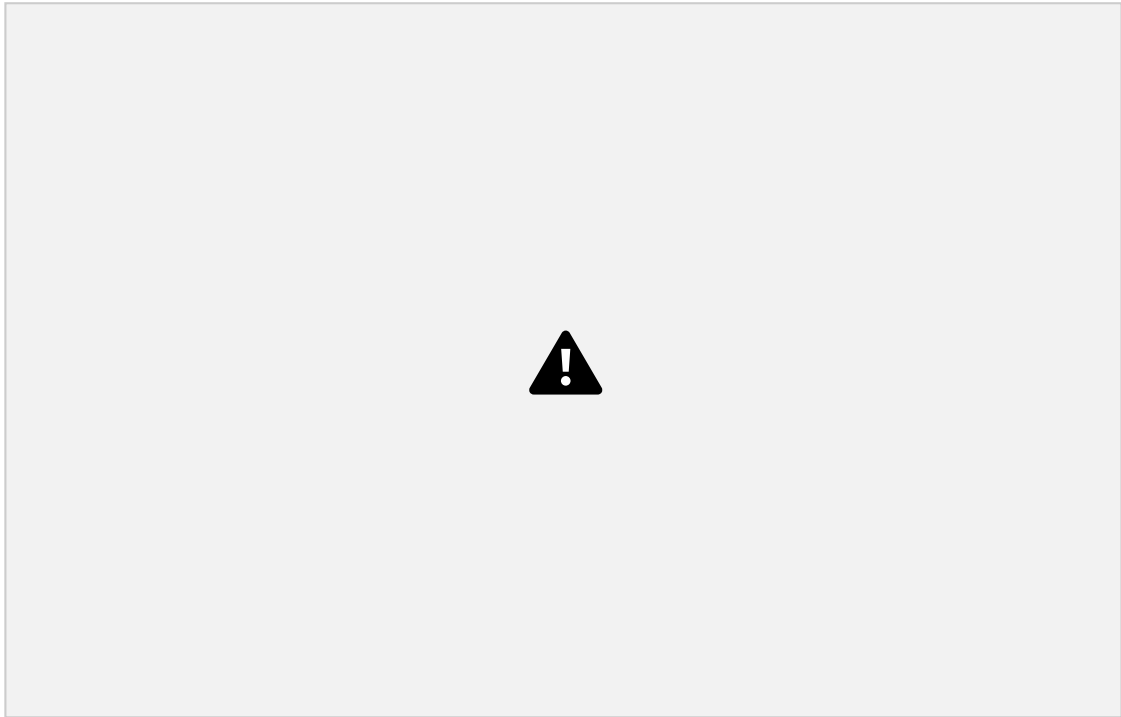
**The step to create the dataset are as follows:**

**Step 1:** Open Microsoft Excel and click on Save As. Select Other .Formats



**Step 2:**

Enter the name of the dataset and Save the dataset astye CSV(MS-DOS).



**Software**

**Step 3:** Enter the name of features as column header.



**Step 3:** Fill the data by using **RANDBETWEEN** function. For every feature , fill the data by considering above specified range. one example is given:



Scroll down the cursor for 30 rows to create 30 instances.

Repeat this for the features, Reading\_Score, Writing\_Score, Placement\_Score, Club\_Join\_Date.

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The placement count largely depends on the placement score. It is considered that if placement score  $<75$ , 1 offer is facilitated; for placement score  $>75$ , 2 offer is facilitated and for else ( $>85$ ) 3 offer is facilitated. Nested If formula is used for ease of data filling.



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**Step 4:** In 20% data, fill the impurities. The range of math score is  $[60,80]$ , updating a few instances values below 60 or above 80. Repeat this for Writing\_Score  $[60,80]$ , Placement\_Score $[75-100]$ , Club\_Join\_Date  $[2018-2021]$ .



**Step 5:**

To violate the rule of response variable, update few values. If placement score is greater than 85, facilitated only 1 offer.



The dataset is created with the given description.

## 2. Identification and Handling of Null Values

Missing Data can occur when no information is provided for one or more items or for a whole unit. Missing Data is a very big problem in real-life scenarios. Missing Data can also refer to as NA (Not Available) values in pandas. In DataFrame sometimes many datasets simply arrive with missing data, either because it exists and was not collected or it never existed. For Example, Suppose different users being surveyed may choose not to share their income, some users may choose not to share the address in this way many datasets went missing.

In Pandas missing data is represented by two values:

1. **None**: None is a Python singleton object that is often used for missing data in Python code.
2. **NaN**: NaN (an acronym for Not a Number), is a special floating-point value recognized by all systems that use the standard IEEE floating-point representation.

Pandas treat None and NaN as essentially interchangeable for indicating missing or null values. To facilitate this convention, there are several useful functions for detecting, removing, and replacing null values in Pandas DataFrame :

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- `isnull()`
- `notnull()`
- `dropna()`
- `fillna()`



- replace()

## 1. Checking for missing values using isnull() and notnull()

### • Checking for missing values using isnull()

In order to check null values in Pandas DataFrame, isnull() function is used. This function return dataframe of Boolean values which are True for NaN values.

#### Algorithm:

**Step 1 :** Import pandas and numpy in order to check missing values in Pandas

DataFrame import pandas as pd import numpy as np

**Step 2:** Load the dataset in dataframe object df

df=pd.read\_csv("/content/StudentsPerformanceTest1.csv")

**Step 3:** Display the data frame

**df**



**Step 4:** Use isnull() function to check null values in the dataset.

df.isnull()



**Step 5:** To create a series true for NaN values for specific columns. for example, math score in dataset and display data with only math score as NaN

```
series = pd.isnull(df["math score"]) df[series]
```

#### · Checking for missing values using notnull()

In order to check null values in Pandas Dataframe, notnull() function is used. This function return dataframe of Boolean values which are False for NaN values.

#### Algorithm:

**Step 1 :** Import pandas and numpy in order to check missing values in Pandas

```
DataFrame import pandas as pd import numpy as np
```

**Step 2:** Load the dataset in dataframe object df

```
df=pd.read_csv("/content/StudentsPerformanceTest1.csv")
```

**Step 3:** Display the data frame

```
df
```

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**Step 4:** Use notnull() function to check null values in the dataset.

```
df.notnull()
```



**Step 5:** To create a series true for NaN values for specific columns. for example, math score in dataset and display data with only math score as NaN

```
series1 = pd.notnull(df["math score"]) df[series1]
```



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See that there are also categorical values in the dataset, for this, you need to use Label Encoding or One Hot Encoding.

- `from sklearn.preprocessing import LabelEncoder`
- `le = LabelEncoder()`
- `df['gender'] = le.fit_transform(df['gender'])`
- `newdf=df df`



## 2. Filling missing values using `dropna()`, `fillna()`, `replace()`

In order to fill null values in a datasets, `fillna()`, `replace()` functions are used. These functions replace NaN values with some value of their own. All these functions help in filling null values in datasets of a DataFrame.

- **For replacing null values with NaN**

```
missing_values = ["Na", "na"]
```

```
df = pd.read_csv("StudentsPerformanceTest1.csv", na_values = missing_values)
```

```
df
```



- **Filling null values with a single value**

**Step 1 :** Import pandas and numpy in order to check missing values in Pandas

```
DataFrame import pandas as pd import numpy as np
```

**Step 2:** Load the dataset in dataframe object df

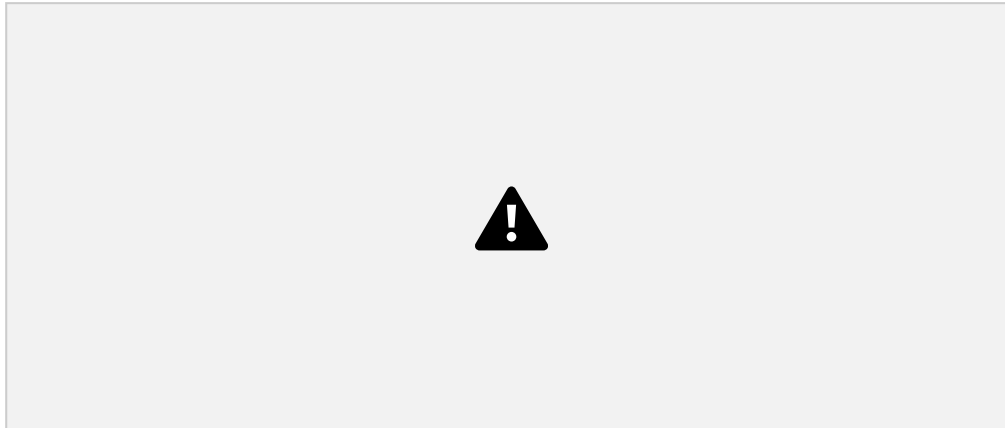
```
df=pd.read_csv("/content/StudentsPerformanceTest1.csv")
```

**Step 3:** Display the data frame

**df**

**Step 4:** filling missing value using fillna()

```
ndf=df.ndf.fillna(0)
```



**Step 5:** Filling missing values using mean, median and standard deviation of that column.

```
data['math score'] = data['math score'].fillna(data['math  
score'].mean())
```

```
data["math score"] = data["math score"].fillna(data["math score"].median())
```

```
data['math score'] = data['math score'].fillna(data["math score"].std())
```

Replacing missing values in forenoon column with minimum/maximum number of that

```
column data["math score"] = data["math score"].fillna(data["math score"].min())
```

```
data["math score"] = data["math score"].fillna(data["math score"].max())
```

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- **Filling null values in dataset**

To fill null values in dataset use inplace=true

```
m_v=df['math score'].mean()
```

```
df['math score'].fillna(value=m_v, inplace=True) df
```



- **Filling a null values using replace () method**

Following line will replace Nan value in dataframe with value -99

```
ndf.replace(to_replace = np.nan, value = -99)
```



- **Deleting null values using dropna() method**

In order to drop null values from a dataframe, dropna() function is used. This function drops Rows/Columns of datasets with Null values in different ways.

1. Dropping rows with at least 1 null value

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2. Dropping rows if all values in that row are missing
3. Dropping columns with at least 1 null value.
4. Dropping Rows with at least 1 null value in CSV file

**Algorithm:**

**Step 1 :** Import pandas and numpy in order to check missing values in Pandas

DataFrame import pandas as pd import numpy as np

**Step 2:** Load the dataset in dataframe object df

```
df=pd.read_csv("/content/StudentsPerformanceTest1.csv")
```

**Step 3:** Display the data frame

**df**

**Step 4:**To drop rows with at least 1 null value

```
ndf.dropna()
```



**Step 5:** To Drop rows if all values in that row are missing

```
ndf.dropna(how = 'all')
```



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**Step 6:** To Drop columns with at least 1 null value.

```
ndf.dropna(axis = 1)
```



**Step 7 :** To drop rows with at least 1 null value in CSV file. making new data frame with dropped NA values

```
new_data = ndf.dropna(axis = 0, how ='any') new_data
```



### 3. Identification and Handling of Outliers

#### 3.1 Identification of Outliers

One of the most important steps as part of data preprocessing is detecting and treating the outliers as they can negatively affect the statistical analysis and the training process of a machine learning algorithm resulting in lower accuracy.

##### 1. What are Outliers?

We all have heard of the idiom 'odd one out' which means something unusual in comparison to the others in a group. Similarly, an Outlier is an observation in a given dataset that lies far from the rest of the observations. That means an outlier is vastly larger or smaller than the remaining values in the set.

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##### 2. Why do they occur?

An outlier may occur due to the variability in the data, or due to experimental error/human error. They may indicate an experimental error or heavy skewness in the data (heavy-tailed distribution).

##### 3. What do they affect?

In statistics, we have three measures of central tendency namely Mean, Median, and Mode. They help us describe the data.

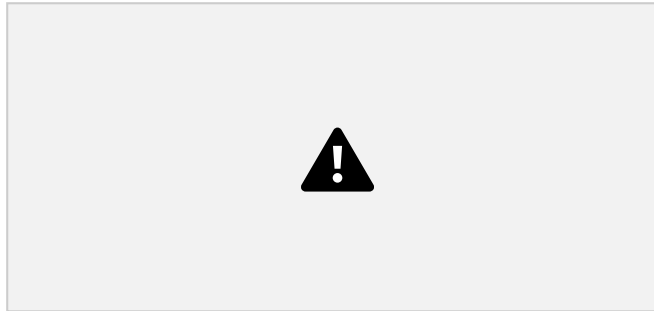
Mean is the accurate measure to describe the data when we do not have any outliers present. Median is used if there is an outlier in the dataset. Mode is used if there is an outlier AND about  $\frac{1}{2}$  or more of the data is the same.

'Mean' is the only measure of central tendency that is affected by the outliers which in turn impacts Standard deviation.



### ■ Example:

Consider a small dataset, sample= [15, 101, 18, 7, 13, 16, 11, 21, 5, 15, 10, 9]. By looking at it, one can quickly say '101' is an outlier that is much larger than the other values.



*fig. Computation with and without outlier*

From the above calculations, we can clearly say the Mean is more affected than the Median.

## 4. Detecting Outliers

If our dataset is small, we can detect the outlier by just looking at the dataset. But what if we have a huge dataset, how do we identify the outliers then? We need to use visualization and mathematical techniques.

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Below are some of the techniques of detecting outliers

- Boxplots
- Scatterplots
- Z-score
- Inter Quantile Range(IQR)

### 4.1 Detecting outliers using Boxplot:

It captures the summary of the data effectively and efficiently with only a simple box and whiskers. Boxplot summarizes sample data using 25th, 50th, and 75th percentiles. One can just get insights(quantiles, median, and outliers) into the dataset by just looking at its boxplot.

### Algorithm:

**Step 1 :** Import pandas and numpy libraries

import pandas as pd  
import numpy as np

**Step 2:** Load the dataset in dataframe object df

```
df=pd.read_csv("/content/demo.csv")
```

**Step 3:** Display the data frame

**df**



**Step 4:** Select the columns for boxplot and draw the boxplot.

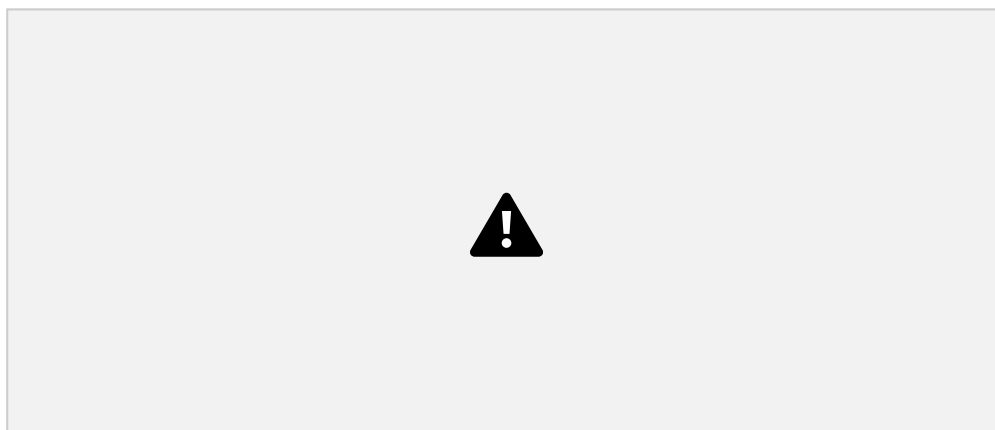
```
col = ['math score', 'reading score', 'writing score','placement score']  
df.boxplot(col)
```

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**Step 5:** We can now print the outliers for each column with reference to the box plot.

```
print(np.where(df['math score']>90)) print (np.where (df ['reading score']< 25))  
print(np.where(df['writing score']<30))
```

#### **4.2 Detecting outliers using Scatterplot:**

It is used when you have paired numerical data, or when your dependent variable has

multiple values for each reading independent variable, or when trying to determine the relationship between the two variables. In the process of utilizing the scatter plot, one can also use it for outlier detection.

To plot the scatter plot one requires two variables that are somehow related to each other. So here Placement score and Placement count features are used.

### **Algorithm:**

Step 1 : Import pandas , numpy and matplotlib libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Load the dataset in dataframe object df

```
df=pd.read_csv("/content/demo.csv")
```

Step 3: Display the data frame

```
df
```

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Step 4: Draw the scatter plot with placement score and placement offer count fig, ax =  
plt.subplots(figsize = (18,10)) ax.scatter(df['placement score'], df['placement offer count'])  
plt.show()

Labels to the axis can be assigned (Optional)

```
ax.set_xlabel('(Proportion non-retail business acres)/(town)')
```

```
ax.set_ylabel('(Full-value property-tax rate)/($10,000)')
```



**Step 5:** We can now print the outliers with reference to scatter plot.

```
print(np.where((df['placement score']<50) & (df['placement offer count']>1)))  
print(np.where((df['placement score']>85) & (df['placement offer count']<3)))
```

#### 4.3 Detecting outliers using Z-Score:

Z-Score is also called a standard score. This value/score helps to understand how far is the data point from the mean. And after setting up a threshold value one can utilize z score values of data points to define the outliers.

$$\text{Zscore} = (\text{data\_point} - \text{mean}) / \text{std. deviation}$$

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#### **Algorithm:**

**Step 1 :** Import numpy and stats from scipy libraries

```
import numpy as np  
from scipy import stats
```

**Step 2:** Calculate Z-Score for mathscore column

```
z = np.abs(stats.zscore(df['math score']))
```

**Step 3:** Print Z-Score Value. It prints the z-score values of each data item of the column

```
print(z)
```



**Step 4:** Now to define an outlier threshold value is chosen.

threshold = 0.18

**Step 5:** Display the sample outliers

```
sample_outliers = np.where(z < threshold) sample_outliers
```

#### 4.4 Detecting outliers using Inter Quartile Range(IQR):

IQR (Inter Quartile Range) Inter Quartile Range approach to finding the outliers is the most commonly used and most trusted approach used in the research field.

$IQR = \text{Quartile3} - \text{Quartile1}$

To define the outlier base value is defined above and below datasets normal range namely Upper and Lower bounds, define the upper and the lower bound ( $1.5 * IQR$  value is considered) :

$\text{upper} = Q3 + 1.5 * IQR$   $\text{lower} = Q1 - 1.5 * IQR$

In the above formula as according to statistics, the 0.5 scale-up of IQR ( $\text{new\_IQR} = IQR + 0.5 * IQR$ ) is taken.

##### Algorithm:

**Step 1 :** Import numpy library

```
import numpy as np
```

**Step 2:** Sort Reading Score feature and store it into sorted\_rscore.

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```
sorted_rscore= sorted(df['reading score'])
```

**Step 3:** Print sorted\_rscore

```
sorted_rscore
```

**Step 4:** Calculate and print Quartile 1 and Quartile 3

```
q1 = np.percentile(sorted_rscore, 25)
```

```
q3 = np.percentile(sorted_rscore, 75) print(q1,q3)
```

**Step 5:** Calculate value of IQR (Inter Quartile Range)

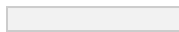
$$\text{IQR} = q3 - q1$$

**Step 6:** Calculate and print Upper and Lower Bound to define the outlier base value. `lwr_bound = q1-(1.5*IQR)` `upr_bound = q3+(1.5*IQR)`  
`print(lwr_bound, upr_bound)`



**Step 7:** Print Outliers

```
r_outliers = []  
for i in sorted_rscore:  
    if (i<lwr_bound or i>upr_bound): r_outliers.append(i)  
print(r_outliers)
```



### 3.2 Handling of Outliers:

For removing the outlier, one must follow the same process of removing an entry from the dataset using its exact position in the dataset because in all the above methods of detecting the outliers end result is the list of all those data items that satisfy the outlier definition according to the method used.

Below are some of the methods of treating the outliers

- Trimming/removing the outlier
- Quantile based flooring and capping
- Mean/Median imputation

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- **Trimming/removing the outlier:**

In this technique, we remove the outliers from the dataset. Although it is not a good practice to follow.

```
new_df=df
```

```
for i in sample_outliers: new_df.drop(i,inplace=True)
```

```
new_df
```



Here Sample outliers are  So instances with index 0, 12, 16 and 17 are deleted.

- **Quantile based flooring and capping:**

In this technique, the outlier is capped at a certain value above the 90th percentile value or floored at a factor below the 10th percentile value

```
df=pd.read_csv("/demo.csv")
df_stud=df
ninetieth_percentile = np.percentile(df_stud['math score'], 90)
b = np.where(df_stud['math score']>ninetieth_percentile, ninetieth_percentile, df_stud['math score'])
print("New array:",b)
```

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```
df_stud.insert(1,"mscore",b,True) df_stud
```



- **Mean/Median Imputation:**

As the mean value is highly influenced by the outliers, it is advised to replace the outliers with the median value.

1. Plot the box plot for reading score `col = ['reading score'] df.boxplot(col)`



2. Outliers are seen in box plot.
3. Calculate the median of reading score by using `sorted_rscore`  
`median=np.median(sorted_rscore) median`
4. Replace the upper bound outliers using median value

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```
refined_df=df
```

```
refined_df['reading score'] = np.where(refined_df['reading score'] > upr_bound,  
median, refined_df['reading score'])
```

5. Display redefined\_df





1. Replace the

lower bound outliers using median value `refined_df['reading score'] =`

`np.where(refined_df['reading score'] < lwr_bound, median, refined_df['reading score'])` 2. Display redefined\_df



3. Draw the box plot for redefined\_df col = ['reading score'] `refined_df.boxplot(col)`





#### 4. Data Transformation for the purpose of:

Data transformation is the process of converting raw data into a format or structure that would be more suitable for model building and also data discovery in general. The process of data transformation can also be referred to as extract/transform/load (ETL). The extraction phase involves identifying and pulling data from the various source systems that create data and then moving the data to a single repository. Next, the raw data is cleansed, if needed. It's then transformed into a target format that can be fed into operational systems or into a data warehouse, a data lake or another repository for use in business intelligence and analytics applications. The transformation The data are transformed in ways that are ideal for mining the data. The data transformation involves steps that are.

- **Smoothing:** It is a process that is used to remove noise from the dataset using some algorithms. It allows for highlighting important features present in the dataset. It helps in predicting the patterns ●

**Aggregation:** Data collection or aggregation is the method of storing and presenting data in a summary format. The data may be obtained from multiple data sources to integrate these data sources into a data analysis description. This is a crucial step since the accuracy of data analysis insights is highly dependent on the quantity and quality of the data used.

- **Generalization:** It converts low-level data attributes to high-level data attributes using concept hierarchy. For Example, Age initially in Numerical form (22, 25) is converted into categorical value (young, old).

- **Normalization:** Data normalization involves converting all data variables into a given range.

Some of the techniques that are used for accomplishing normalization are:

- **Min-max normalization:** This transforms the original data linearly.

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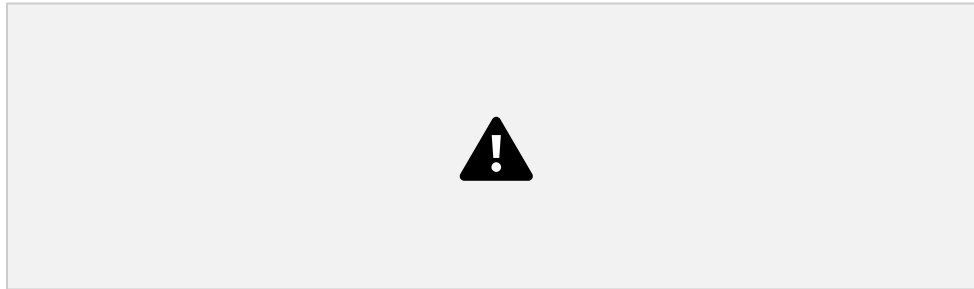
- **Z-score normalization:** In z-score normalization (or zero-mean normalization) the

values of an attribute (A), are normalized based on the mean of A and its standard deviation.

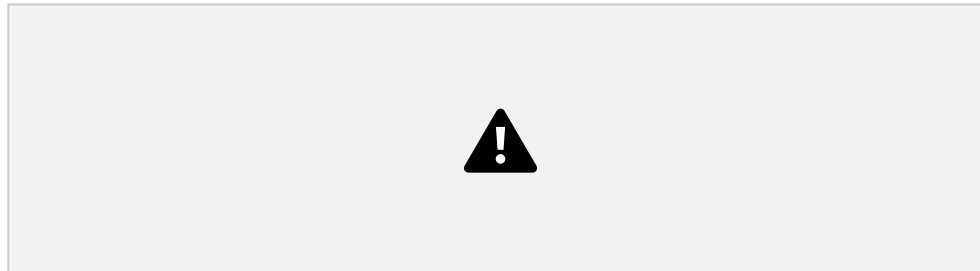
- **Normalization by decimal scaling:** It normalizes the values of an attribute by changing the position of their decimal points

- **Attribute or feature construction.**

- **New attributes constructed from the given ones:** Where new attributes are created & applied to assist the mining process from the given set of attributes. This simplifies the original data & makes the mining more efficient.



a. To change the scale for better understanding (Attribute or feature construction)  
Here the Club\_Join\_Date is transferred to Duration.



### Algorithm:

**Step 1 :** Import pandas and numpy libraries

```
import pandas as pd  
import numpy as np
```

**Step 2:** Load the dataset in dataframe object df

```
df=pd.read_csv("/content/demo.csv")
```

**Step 3:** Display the data frame

```
df
```

**Step 4:** Change the scale of Joining year to duration.

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b. To decrease the skewness and convert distribution into normal distribution

## (Normalization by decimal scaling)

**Data Skewness:** It is asymmetry in a statistical distribution, in which the curve appears distorted or skewed either to the left or to the right. Skewness can be quantified to define the extent to which a distribution differs from a normal distribution.

**Normal Distribution:** In a normal distribution, the graph appears as a classical, symmetrical “bell-shaped curve.” The mean, or average, and the mode, or maximum point on the curve, are equal.

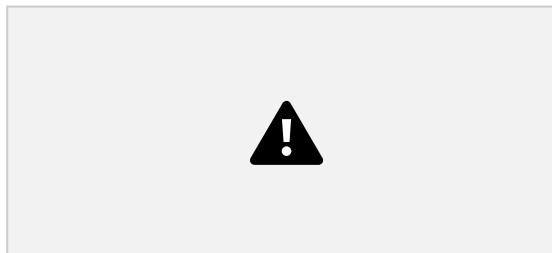
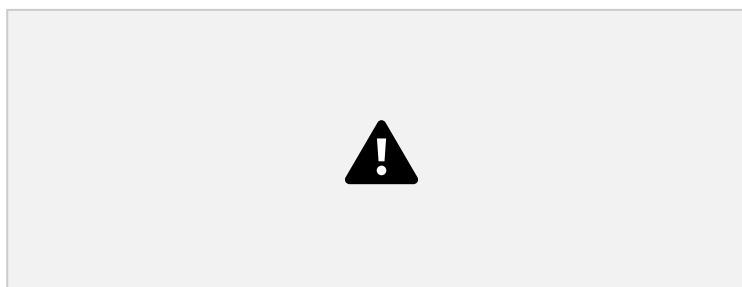


Fig. Positively Skewed Distribution

**A positively skewed distribution** means that the extreme data results are larger. This skews the data in that it brings the mean (average) up. The mean will be larger than the median in a Positively skewed distribution.

**A negatively skewed distribution** means the opposite: that the extreme data results are smaller. This means that the mean is brought down, and the median is larger than the mean in a negatively skewed distribution.



**Reducing skewness** A data transformation may be used to reduce skewness. A distribution that is symmetric or nearly so is often easier to handle and interpret than a skewed distribution. The logarithm,  $x$  to log base 10 of  $x$ , or  $x$  to log base  $e$  of  $x$  ( $\ln x$ ), or  $x$  to log base 2 of  $x$ , is a strong

transformation with a major effect on distribution shape. It is commonly used for reducing right skewness and is often appropriate for measured variables. It can not be applied to zero or negative values.



### Algorithm:

**Step 1 :** Detecting outliers using Z-Score for the Math\_score variable and remove the outliers.

**Step 2:** Observe the histogram for math\_score variable. import matplotlib.pyplot as plt  
new\_df['math score'].plot(kind = 'hist')

**Step 3:** Convert the variables to logarithm at the scale 10.

df['log\_math'] = np.log10(df['math score'])

**Step 4:** Observe the histogram for math\_score variable.

df['log\_math'].plot(kind = 'hist')



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### Conclusion:

In this way we have explored the functions of the python library for Data Identifying and handling the outliers. Data Transformations Techniques are explored with the purpose of creating the new variable and reducing the skewness from datasets.

**Via Question:**

1. Explain the methods to detect the outlier.
2. Explain data transformation methods
3. Write the algorithm to display the statistics of Null values present in the dataset.
4. Write an algorithm to replace the outlier value with the mean of the variable.

**Coding**

**Completion**

**Total Dated Sign of Course In-charge**

**Efficiency Viva Timely**

**5 3 2 10**

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**Experiment No: 3**

## Descriptive Statistics - Measures of Central Tendency and Variability

**Name of the Student:**

\_\_\_\_\_ **Class: Roll No.:**

**Batch:**

Date: Mark: /10

**Signature of the Course In-charge:** \_\_\_\_\_

**Signature of the HOD:**

\_\_\_\_\_

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**Group:A**

**Assignment No: 3**

**Title of the Assignment:** Descriptive Statistics - Measures of Central Tendency and Variability

Perform the following operations on any open source dataset (e.g., data.csv)

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a

dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variables. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.

2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

Provide the codes with outputs and explain everything that you do in this step.

**Objective:** Students should be able to perform the Statistical operations using Python on any open source dataset.

**Prerequisite:**

1. Basic of Python Programming
2. Concept of statistics such as mean, median, minimum, maximum, standard deviation etc.

**Software Requirements:**

- Anaconda with Python 3.7

**Hardware Requirement:**

- PIV, 2GB RAM, 500 GB HDD.

**Outcome:**

Students understand and apply descriptive statistics by calculating measures of central tendency and variability on real-world datasets. It also enables learners to write Python code for generating statistical summaries and analysing grouped data effectively.

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**Theory:**

**What is Statistics?**

Statistics is the science of collecting data and analysing them to infer proportions (sample) that are representative of the population. In other words, statistics is interpreting data in order to make



predictions for the population.

## Branches of Statistics:

There are two branches of Statistics.

**1. Descriptive Statistics:** Descriptive Statistics is a statistics or a measure that describes the data.

**2. Inferential Statistics:** Using a random sample of data taken from a population to describe and make inferences about the population is called Inferential Statistics.

## Descriptive Statistics

Descriptive Statistics is summarizing the data at hand through certain numbers like mean, median etc. so as to make the understanding of the data easier. It does not involve any generalization or inference beyond what is available. This means that the descriptive statistics are just the representation of the data (sample) available and not based on any theory of probability.

## Commonly Used Measures

I. Measures of Central Tendency

II. Measures of Dispersion (or Variability)

### I. Measures of Central Tendency

A Measure of Central Tendency is a one number summary of the data that typically describes the centre of the data. This one number summary is of three types.

#### a. Mean:

Mean is defined as the ratio of the sum of all the observations in the data to the total number of observations. This is also known as Average. Thus mean is a number around which the entire data set is spread. Consider the following data points.

17, 16, 21, 18, 15, 17, 21, 19, 11, 23

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**b. Median :** Median is the point which divides the entire data into two equal halves.

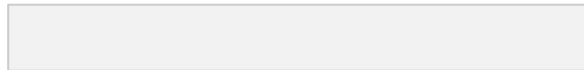
One-half of the data is less than the median, and the other half is greater than the same. Median is calculated by first arranging the data in either ascending or descending order.

- If the number of observations is odd, the median is given by the middle observation in the sorted form.
- If the number of observations are even, median is given by the mean of the two middle observations in the sorted form.
- An important point to note is that the order of the data (ascending or descending) does not affect the median.

To calculate Median, let's arrange the data in ascending order.

11, 15, 16, 17, 17, 18, 19, 21, 21, 23

Since the number of observations is even (10), median is given by the average of the two middle observations (5th and 6th here).



**c. Mode :** Mode is the number which has the maximum frequency in the entire data set, or in other words, mode is the number that appears the maximum number of times. A data can have one or more than one mode.

- If there is only one number that appears the maximum number of times, the data has one mode, and is called Uni-modal.
- If there are two numbers that appear the maximum number of times, the data has two modes, and is called Bi-modal.
- If there are more than two numbers that appear the maximum number of times, the data has more than two modes, and is called Multi-modal.

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Consider the following data points.

17, 16, 21, 18, 15, 17, 21, 19, 11, 23

Mode is given by the number that occurs the maximum number of times. Here, 17 and 21 both occur twice. Hence, this is a Bimodal data and the modes are 17 and 21.

## **II. Measures of Dispersion (or Variability)**

Measures of Dispersion describes the spread of the data around the central value (or the Measures of Central Tendency)

**a. Absolute Deviation from Mean** — The Absolute Deviation from Mean, also called Mean Absolute Deviation (MAD), describes the variation in the data set, in the sense that it tells the average absolute distance of each data point in the set. It is calculated as

$$\text{MAD} = \frac{\sum |x_i - \bar{x}|}{n}$$

**b. Variance** — Variance measures how far are data points spread out from the mean. A high variance indicates that data points are spread widely and a small variance indicates that the data points are closer to the mean of the data set. It is calculated as

$$\text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n}$$

**c. Standard Deviation** — The square root of Variance is called the Standard Deviation. It is calculated as

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

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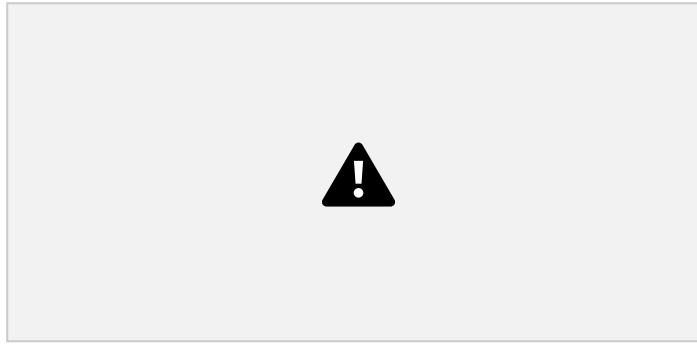
**d. Range** — Range is the difference between the Maximum value and the Minimum value in the data set. It is given as

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

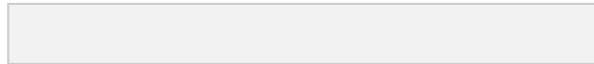
**e. Quartiles** — Quartiles are the points in the data set that divides the data set into four equal parts. Q1, Q2 and Q3 are the first, second and third quartile of the data set.

i. 25% of the data points lie below Q1 and 75% lie above it.

- ii. 50% of the data points lie below Q2 and 50% lie above it. Q2 is nothing but Median.
- iii. 75% of the data points lie below Q3 and 25% lie above it.



**f. Skewness** — The measure of asymmetry in a probability distribution is defined by Skewness. It can either be positive, negative or undefined.



**Positive Skew** — This is the case when the tail on the right side of the curve is bigger than that on the left side. For these distributions, mean is greater than the mode.

**Negative Skew** — This is the case when the tail on the left side of the curve is bigger than that on the right side. For these distributions, mean is smaller than the mode.

The most commonly used method of calculating Skewness is

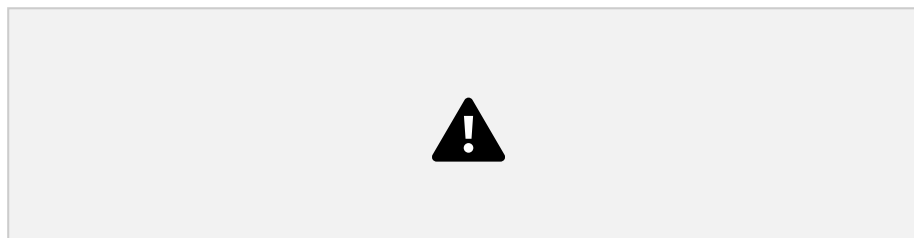
If the skewness is zero, the distribution is symmetrical. If it is negative, the distribution is Negatively Skewed and if it is positive, it is Positively Skewed.

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**Python Code:**

**1. Mean** To find mean of all columns

Syntax:

df.mean()

tput:



To find mean of specific column

Syntax:

df.loc[:, 'Age'].mean() 38.85

To find mean row wise

Syntax:

df.mean(axis=1)[0:4]

ut:



df.mode()

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## 1. Median

**To find median of all columns**

Syntax:

df.median()



To find median of specific column

Syntax:

df.loc[:, 'Age'].median()

t:

**Output:** 36.0

To find median row wise

Syntax:

```
df.median(axis=1)[0:4]
```



## 2. Mode

To find mode of all columns

Syntax:



t:

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In the Genre Column mode is Female, for column Age mode is 32 etc. If a particular column does not have mode all the values will be displayed in the column.

To find the mode of a specific column. Syntax:

```
df.loc[:, 'Age'].mode()
```

ut:

Output: 32

## 3. Minimum

**To find minimum of all columns**

Syntax:

```
df.min()
```



To find minimum of Specific column

Syntax:

```
df.loc[:, 'Age'].min(skipna = False)
```

Output: **18**

## 1. Maximum

**To find Maximum of all columns**

Syntax:

```
df.max()
```

t:



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**To find Maximum of Specific**

**column Syntax:**

```
df.loc[:, 'Age'].max(skipna =
```

Output:

```
False) 70
```

## 2. Standard Deviation

To find Standard Deviation of all columns

Syntax:

```
df.std()
```



To find Standard Deviation of specific column

Syntax:

```
df.loc[:, 'Age'].std()
```

**Output:** 13.969007331558883

A variable is a characteristic that can be measured and that can assume different values. Height, age, income, province or country of birth, grades obtained at school and type of housing are all examples of variables. Variables may be classified into two main categories:

- Categorical and
- Numeric.

Each category is then classified in two subcategories: nominal or ordinal for categorical variables, discrete or continuous for numeric variables.

put:  
variabl

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### • **Categorical Variables**

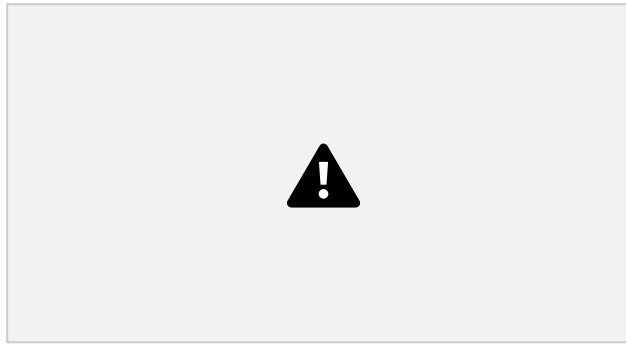
A categorical variable (also called qualitative variable) refers to a characteristic that can't be quantifiable.

Categorical variables can be either nominal or ordinal.

#### • **Nominal Variable**

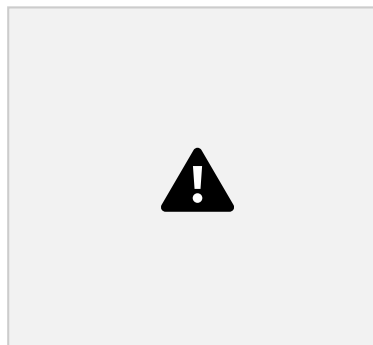
A nominal variable is one that describes a name, label or category without natural order. In the given table, the variable “mode of transportation for travel to work” is also nominal.





- **Ordinal Variable**

An ordinal variable is a variable whose values are defined by an order relation between the different categories. In the following table, the variable “behaviour” is ordinal because the category “Excellent” is better than the category “Very good,” which is better than the category “Good,” etc. There is some natural ordering, but it is limited since we do not know by how much “Excellent” behaviour is better than “Very good” behaviour.



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- **Numerical Variables**

A numeric variable (also called quantitative variable) is a quantifiable characteristic whose values are numbers (except numbers which are codes standing up for categories). Numeric variables may be either continuous or discrete.

- **Continuous variables**

A variable is said to be continuous if it can assume an infinite number of real values within a given interval.

For instance, consider the height of a student. The height can't take any values. It can't be negative and it can't be higher than three metres. But between 0 and 3, the number of possible

values is theoretically infinite. A student may be 1.6321748755 ... metres tall.

### ▪ Discrete variables

As opposed to a continuous variable, a discrete variable can assume only a finite number of real values within a given interval.

An example of a discrete variable would be the score given by a judge to a gymnast in competition: the range is 0 to 10 and the score is always given to one decimal.

## 2. Summary statistics of income grouped by the age groups

**Problem Statement:** For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable. Categorical Variable: Genre Quantitative Variable: Age Syntax:

```
df.groupby(['Genre'])['Age'].mean() Output:
```



Categorical Variable: Genre Quantitative Variable : Income Syntax:

```
df_u=df.rename(columns= {'Annual Income k$':'Income'},inplace=False)
```

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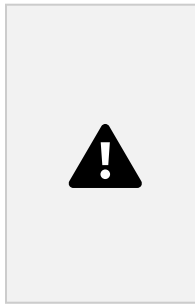
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```
(df_u.groupby(['Genre']).Income.mean()) Output:
```



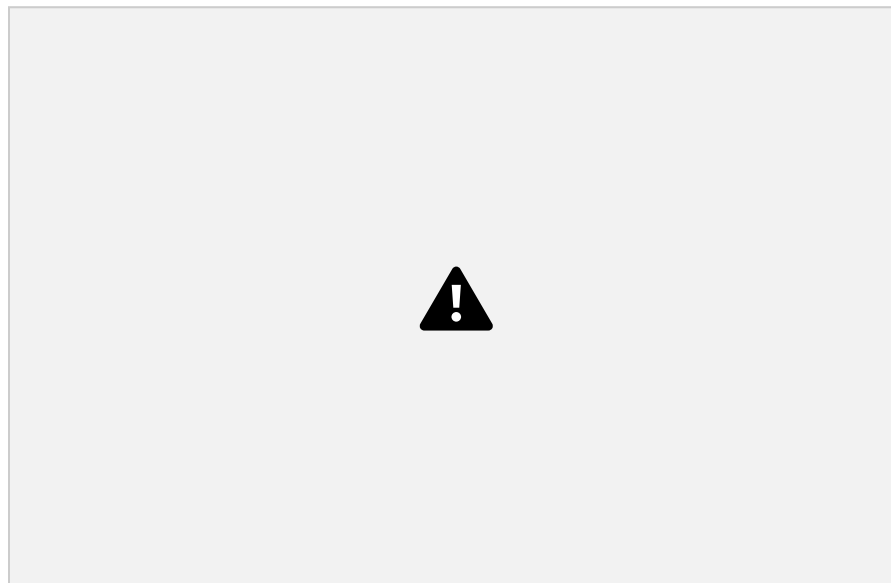
To create a list that contains a numeric value for each response to the categorical variable. from sklearn import preprocessing enc = preprocessing.OneHotEncoder()

```
enc_df = pd.DataFrame(enc.fit_transform(df[['Genre']]).toarray()) enc_df
```



To concat numerical list to dataframe `df_encode=df_u.join(enc_df)`

`df_encode`



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### **3. Display basic statistical details on the iris dataset. Algorithm:**

1. Import Pandas Library
2. The dataset is downloaded from UCI repository.

`csv_url =`

`'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'` 3. Assign

Column names

`col_names = ['Sepal_Length','Sepal_Width','Petal_Length','Petal_Width','Species']`

4. Load Iris.csv into a Pandas data frame

```
iris = pd.read_csv(csv_url, names = col_names)
```

5. Load all rows with Iris-setosa species in variable irisSet

```
irisSet = (iris['Species']== 'Iris-setosa')
```

6. To display basic statistical details like percentile, mean, standard deviation etc. for Iris setosa use describe

```
print('Iris-setosa') print(iris[irisSet].describe())
```

7. Load all rows with Iris-versicolor species in variable irisVer

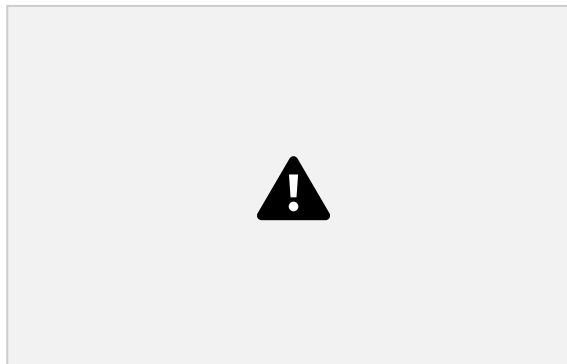
```
irisVer = (iris['Species']== 'Iris-versicolor')
```

8. To display basic statistical details like percentile, mean, standard deviation etc. for Iris versicolor use describe

```
print('Iris-versicolor') print(iris[irisVer].describe())
```

9. Load all rows with Iris-virginica species in variable irisVir

```
irisVir = (iris['Species']== 'Iris-virginica')
```



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### **Conclusion:**

Descriptive statistics summarises or describes the characteristics of a data set. Descriptive statistics consists of two basic categories of measures:

- measures of central tendency and
- measures of variability (or spread).

Measures of central tendency describe the centre of a data set. It includes the mean, median,

and mode.

Measures of variability or spread describe the dispersion of data within the set and it includes standard deviation, variance, minimum and maximum variables.

**Via Questions:**

1. Explain Measures of Central Tendency with examples.
2. What are the different types of variables? Explain with examples.
3. Which method is used to statistic the data frame? write the code.

**Coding**

**Completion**

**TotalDated Sign of Course In-charge**

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**5 3 2 10**

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**Experiment No: 4**

## Data Analytics - I

**Name of the Student:**

\_\_\_\_\_ **Class: Roll No.:**

**Batch:**

Date: Mark: /10

**Signature of the Course In-charge:** \_\_\_\_\_

**Signature of the HOD:**

\_\_\_\_\_

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**Group A**

**Assignment No: 4**

**Title of the Assignment:** Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (<https://www.kaggle.com/c/boston-housing>). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.

**Objective:** Students should be able to data analysis using liner regression using Python for any open source dataset.

**Prerequisite:**

1. Basic of Python Programming
2. Concept of Regresion.

**Software Requirements:**

- Anaconda with Python 3.7

**Hardware Requirement:**

- PIV, 2GB RAM, 500 GB HDD.

**Outcome:**

Students will be able to build and evaluate a Linear Regression model using Python to predict house prices using the Boston Housing dataset. They will understand data pre-processing, model training, performance evaluation, and interpretation of key features affecting housing prices.

**Theory:**

**Linear Regression:** It is a machine learning algorithm based on supervised learning. It targets prediction values on the basis of independent variables.

- It is preferred to find out the relationship between forecasting and variables.
- A linear relationship between a dependent variable (X) is continuous; while independent variable(Y) relationship may be continuous or discrete. A linear relationship should be available in between predictor and target variable so known as Linear Regression.
- Linear regression is popular because the cost function is Mean Squared Error (MSE) which is

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~~equal to the average squared difference between an observation's actual and predicted values.~~

- It is shown as an equation of line like :  $Y = m \cdot X + b + e$

Where: b is intercepted, m is slope of the line and e is error term.

This equation can be used to predict the value of target variable Y based on given predictor variable(s) X, as shown in Fig. 1.

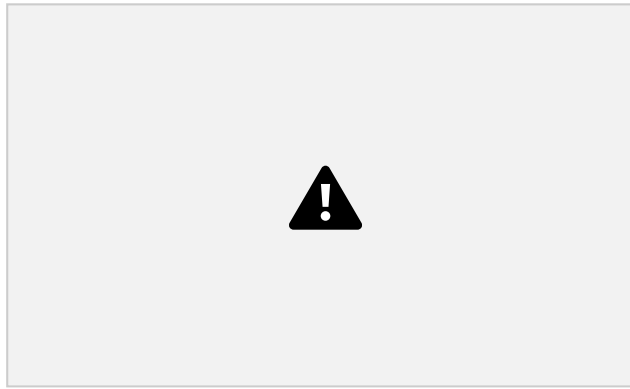


Fig. 1: geometry of linear regression

- Fig. 2 shown below is about the relation between weight (in Kg) and height (in cm), a linear relation. It is an approach of studying in a statistical manner to summarise and learn the relationships among continuous (quantitative) variables.
- Here a variable, denoted by 'x' is considered as the predictor, explanatory, or independent variable.
- Another variable, denoted 'y', is considered as the response, outcome, or dependent variable. While "predictor" and "response" used to refer to these variables.
- Simple linear regression technique concerned with the study of only one predictor variable.

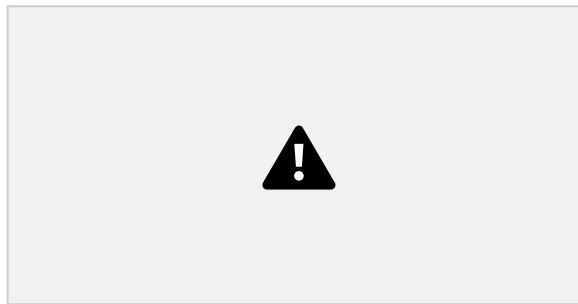


Fig.2 : Relation between weight (in Kg) and height (in cm)